

Department of Forest Resources Management
University of British Columbia
2424 Main Mall
Vancouver, BC V6T 1Z4
Canada

October 27, 2025

Editor-in-Chief
Forest Science

Dear Editor,

Please consider the enclosed manuscript, “A Weighted Fitting Approach for Diameter Distributions from Horizontal Point Sampling,” for publication as a *Brief Communication* in *Forest Science*. The paper introduces a weighted nonlinear least squares estimator that reproduces the classical size-biased fitting procedure while operating directly on horizontal point sampling (HPS) tallies, removing the need to expand samples to stand-table form.

Using provincial inventory data from Québec, I show that the weighted estimator matches the reference size-biased fits to near numerical precision: relative ℓ_2 errors fall below 8.5% in HPS space and 2.4% in stand-table space, and chi-square diagnostics remain within the same order of magnitude across species/cover combinations. The manuscript keeps the presentation concise—one figure and one table—while the accompanying reproducible workflow documents the complete analysis from raw data to manuscript.

All scripts, configuration files, and processed datasets necessary to reproduce the results are openly available in the accompanying GitHub repository (<https://github.com/UBC-FRESH/dbhdistfit-papers>). Because the repository provides the full computational record, no separate supplementary material is submitted.

The work has not been published elsewhere and is not under consideration by another journal. I have no competing interests to declare.

Sincerely,

Gregory E. Paradis
Department of Forest Resources
Management
University of British Columbia
gregory.paradis@ubc.ca